Kesser Karim

AI tools for classrooms · Fairness and reliability

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PROFILE

Builds AI marking and feedback tools for education, including PenBright (formerly MarkMate), an in-house AI marking tool, and FutureTech, a tutoring platform founded for parents in the UAE and the UK. Audits AI for fairness and reliability, including Harbour, which will independently test, monitor and certify AI products used by children (launching soon; to be sold to vendors). Part-time PhD researcher at the University of Portsmouth on fair, calibrated and explainable models for early Type-2 diabetes risk in South Asian populations, with data partnerships at Connected Bradford and UK Biobank. Computer Science teacher for 15 years, from primary through Cambridge IGCSE and A Level, currently at Nord Anglia International School Abu Dhabi. Has led AI education at his school for six years: AI lessons and extra-curricular activities, teacher training, and PenBright. MSc in Computer Science and Data Science.

TECHNICAL SKILLS

Programming: Python (advanced), SQL, R, JavaScript
Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
GenAI and LLM: LangChain, RAG architectures, vector databases (pgvector), evaluation harnesses
ML engineering: FastAPI, Docker, MLflow, Git, GitHub Actions (CI/CD), experiment tracking
Data: pandas, NumPy, ETL and data pipeline design
Visualisation: Matplotlib, Seaborn, Power BI
Cloud: AWS (S3, EC2); Microsoft Azure
Specialist: Clinical risk prediction, fairness auditing (Fairlearn, AIF360), model calibration, explainable AI (SHAP)

DOCTORAL RESEARCH

PhD candidate, Computer Science (part-time) — University of Portsmouth, UK

2024 – present

Research: early detection of Type-2 diabetes risk using AI and machine learning in the South Asian population.

- Developing calibrated, fair and explainable machine learning models for early Type-2 diabetes risk stratification from routinely collected demographic, lifestyle and clinical data.
- Data partnerships with Connected Bradford and UK Biobank; comparative study design across Bradford localities with high and low South Asian population density to assess equity impacts.
- Methods span logistic regression, gradient-boosted trees and deep learning, with calibration analysis (Brier score, reliability diagrams) and subgroup fairness audits (equalised odds, calibration-by-group).
- Conducted part-time alongside a full-time teaching post, so the research has to stay resource-aware and usable in real clinical settings.

TECHNICAL PROJECTS

T2D Risk Predictor — Type-2 diabetes risk on open data

Python, scikit-learn, XGBoost, FastAPI, Docker, MLflow · github.com/kesserk/t2d-risk-predictor · In progress

- Work-in-progress pipeline on the open NHANES dataset, alongside the doctoral research. NHS data is not in this repository. No results are published yet.
- In the repository now: a FastAPI service skeleton, Docker, GitHub Actions CI, an NHANES exploratory notebook and a baseline model notebook.

Diabetes Guidance RAG — questions over NICE NG28 and ADA Standards of Care

Design note only · github.com/kesserk/diabetes-rag · In design

- A planned retrieval-augmented system over NICE NG28 and the ADA Standards of Care. Faithfulness and hallucination checks are part of the design, not an afterthought.
- The repository currently holds that design note only. Ingestion, retrieval, evaluation and serving are not built. This is not a shipped system.

PenBright — AI marking and feedback (formerly MarkMate)

LLM application · Built · penbright.vercel.app

- In-house AI marking tool that generates structured, mark-scheme-aligned feedback, to shorten marking time while holding consistency. Designed around real assessment workflows.

FutureTech — tutoring platform

Founder

- Tutoring platform for parents in the UAE and the UK.

Harbour — independent assurance for AI products used by children

Founder · Launching soon · not yet launched

- Will independently test, monitor and certify AI products used by children, and will be sold to vendors. It has not launched.

Learning analytics early-warning model

Python, scikit-learn, SHAP · Prototype

- Model bringing together NGRT, CAT4, ALIS, TIMSS and internal school assessment data to flag learners who may need support, with explainable reasons and suggested interventions.

PROFESSIONAL EXPERIENCE

Computer Science teacher — Nord Anglia International School Abu Dhabi, UAE

August 2024 – present

- Leads the school's rollout of ADEK's K–12 AI curriculum.
- Played a key role in establishing and growing the Secondary Computer Science department, including resources, schemes of work and digital learning platforms aligned to Cambridge IGCSE and A Level frameworks.
- Led development of a future-focused Computer Science curriculum embedding problem-solving, creativity and digital fluency across key stages.
- Facilitated enrichment through the Nord Anglia–MIT collaboration, guiding students through hands-on STEAM challenges and global innovation projects.
- Founded and lead the school's Minecraft Education Club, using game-based learning to teach computational thinking, teamwork and design thinking.

- Work with the leadership team to promote digital citizenship and safe, purposeful use of technology across the school community.

Gifted & Talented coordinator and Computer Science teacher — Safa Community School, Dubai, UAE

January 2021 – August 2024

- Built Power BI dashboards tracking student performance across approximately 200 students and 6 subjects, replacing reporting workflows previously done manually.
- Automated data ingestion and cleaning pipelines connecting the school management system to visualisation platforms, reducing manual reporting time.
- Designed and delivered a hands-on Python and data science curriculum using Jupyter, covering supervised and unsupervised learning, model evaluation and basic deployment patterns.
- Mentored students on independent machine learning projects including image classification and regression tasks.
- Taught GCSE and A Level Computer Science to students aged 11–18 and contributed to curriculum design across Key Stages 3, 4 and 5.

Mathematics and Computer Science teacher — One in a Million Free School, Bradford, UK

September 2018 – July 2020

- Deputised for the Head of Department, running the Mathematics department in his absence.
- Led the design and planning of a new computing curriculum, including introductory Python, algorithms and data analytics.
- Conducted data analysis of examination performance to identify focus areas, then created and implemented targeted intervention programmes.
- Delivered the Edexcel GCSE Mathematics syllabus across KS3 and KS4; conducted teacher observation, performance management and NQT mentoring.

Computer Science teacher — Lightcliffe Academy, Halifax, UK

March 2018 – July 2018

- Delivered the Computer Science National Curriculum across Years 7–13, introducing the new KS3 curriculum, the OCR GCSE course and a BTEC Level 3 course in ICT.

Curriculum leader, Technology — Trinity Academy Sowerby Bridge, UK

September 2017 – March 2018

- Managed the Technology department covering Computer Science, iMedia and Business Studies, supervising delivery across specialist and non-specialist teachers.
- Implemented intervention strategies and monitored attainment through assessment and target setting.

Computer Science teacher — GEMS Wellington International School, Dubai, UAE

September 2015 – July 2017

- Taught Cambridge IGCSE Computer Science, IGCSE ICT and the IB curriculum to students aged 11–18, and contributed to curriculum design across Key Stages 3 and 4.

Earlier: ICT trainee teacher (PGCE placements) — King James Secondary School, Knaresborough and Bury Church of England High School, UK · 2014–2015

EDUCATION

PhD Computer Science (part-time)

University of Portsmouth, UK · 2024 – present

Early detection of Type-2 diabetes risk using AI and machine learning in the South Asian population

MSc Computer Science and Data Science

University of Sunderland, UK · 2023 – 2024

Machine learning, time series analysis, big data technologies, deep learning with TensorFlow and PyTorch; dissertation applying Bayesian methods to predictive modelling on health-relevant datasets

PGCE

Manchester Metropolitan University, UK · 2014 – 2015

Subject Knowledge Enhancement course, Computing

Manchester Metropolitan University, UK · 2014

LLB Law

Leeds Metropolitan University, UK · 2006 – 2010

PROFESSIONAL DEVELOPMENT AND MEMBERSHIPS

- Member, British Computer Society (BCS).
- DataCamp Machine Learning Scientist with Python track completed; Machine Learning Engineer specialisation in progress.
- Preparing for Microsoft AI-300: Machine Learning Operations (MLOps) Engineer Associate (successor to the retired DP-100).

ADDITIONAL

Languages: English (native), Urdu (basic spoken)
Interests: Football, running and cycling; has run several marathons including London and New York, fundraising for charity
Volunteering: Local tuition centres, supporting students in Years 5–11 with Computer Science and Mathematics